

Regiment Analysis of Multi-Level Organizations

Hierarchy, Incentives, Attrition, and Distributional Outcomes

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Abstract

Multi-level organizations provide a natural empirical environment for Regiment Analysis because individuals are embedded in explicitly hierarchical structures in which recruitment, rewards, performance, promotion, attrition, and network position interact. This paper develops an empirical framework for studying such systems using multi-level marketing organizations as a principal application. The framework treats participants as observations, organizational groups as regiments, and the complete organization as a higher-level network. It studies net financial outcomes, organizational depth, recruitment, retail activity, rewards, sanctions, persistence, and attrition. Particular attention is given to the frequently cited claim that at least 99 percent of recruits in a study of 350 United States multi-level marketing companies lost money. Rather than treating this statistic as an established universal parameter, the paper treats it as an empirical hypothesis requiring replication, appropriate denominators, complete expense accounting, and attention to selection. The framework combines hierarchical statistics, econometrics, network analysis, behavioral economics, organizational psychology, finance, welfare economics, causal inference, machine learning, and dynamic systems analysis. The central objective is to determine how organizational position, incentives, and selection interact to produce observed distributions of gains, losses, and attrition.

The paper ends with “The End”

1 Introduction

Multi-level organizations are systems in which individual participants are embedded within hierarchical and networked structures. Multi-level marketing (MLM) provides a particularly visible example. Participants may sell products, recruit other participants, receive compensation associated with their own activity, and in some organizational designs receive compensation associated with downstream activity.

Such organizations are therefore natural candidates for Regiment Analysis.

The central empirical object is not merely the average participant. It is the distribution of outcomes across organizational positions.

Let participant j belong to organizational regiment i . Define

$$R_i = (x_{i1}, x_{i2}, \dots, x_{iq}). \quad (1)$$

The complete organization consists of

$$\mathcal{R} = \{R_1, R_2, \dots, R_r\}. \quad (2)$$

The analytical objective is to determine how financial outcomes, incentives, recruitment, hierarchy, and attrition vary across these regiments.

2 The Empirical Motivation

A frequently cited claim concerning MLM organizations is that at least 99 percent of recruits lose money. The claim is associated with an analysis of 350 United States MLM companies by Jon Taylor and has subsequently been reproduced in discussions of MLM economics.

The appropriate scientific response is not simply to accept or reject the headline figure. Instead, Regiment Analysis asks:

$$\Pr(\Pi < 0 \mid \mathcal{D}) \quad (3)$$

where Π is participant net financial outcome and $\mathcal{D}$ denotes the relevant organizational and accounting information.

The central empirical problem is therefore to determine the distribution

$$F_{\Pi}(\pi) = \Pr(\Pi \leq \pi). \quad (4)$$

The statistic

$$\Pr(\Pi < 0) \quad (5)$$

is only one feature of that distribution.

3 Why the 99 Percent Statistic Requires Care

Suppose 99 percent of participants lose a small amount while one percent gain extremely large amounts. Then

$$\Pr(\Pi < 0) = 0.99 \quad (6)$$

can coexist with

$$E[\Pi] > 0. \quad (7)$$

Conversely, 99 percent losing money does not reveal whether losses are one dollar, one hundred dollars, or several thousand dollars.

Consequently, a complete analysis should report at least

$$(\Pr(\Pi < 0), E[\Pi], \text{Median}(\Pi), Q_{0.05}(\Pi), Q_{0.95}(\Pi), \text{Var}(\Pi)). \quad (8)$$

It should additionally report the time horizon over which the outcome is measured and the denominator from which participants are sampled.

The distinction is fundamental:

$$\text{loss probability} \neq \text{loss magnitude} \neq \text{expected return}. \quad (9)$$

4 The Regiment Representation

Let i denote an organizational regiment and j an individual participant.

For participant j ,

$$\Pi_{ij} = Y_{ij} - E_{ij}, \quad (10)$$

where Y_{ij} represents monetary receipts and E_{ij} represents economically relevant expenses. A more complete decomposition is

$$\Pi_{ij} = C_{ij} + S_{ij} + B_{ij} - F_{ij} - T_{ij} - M_{ij} - O_{ij}, \quad (11)$$

where:

- C denotes commissions;
- S denotes retail-sales margins;
- B denotes bonuses;
- F denotes fees;
- T denotes training or travel expenses;
- M denotes mandatory or encouraged purchases;
- O denotes other relevant operating costs.

The exact accounting definition must be fixed before estimation.

5 Organizational Hierarchy

Let D_{ij} denote the depth of participant j in the organizational hierarchy.

A participant's organizational position may include:

$$H_{ij} = (D_{ij}, U_{ij}, A_{ij}, R_{ij}), \quad (12)$$

where U is the size of the participant's downstream network, A is organizational ancestry, and R represents recruitment activity.

A basic hypothesis is that financial outcomes may depend on hierarchy:

$$E[\Pi_{ij} \mid D_{ij} = d]. \quad (13)$$

The relationship may be monotonic, nonlinear, or heterogeneous.

Regiment Analysis does not assume the answer.

6 Organizational Depth

Suppose the organization contains levels

$$d = 0, 1, \dots, D. \quad (14)$$

Define the conditional mean

$$\mu_d = E[\Pi \mid D = d]. \quad (15)$$

The empirical depth profile is

$$\mathcal{M} = (\mu_0, \mu_1, \dots, \mu_D). \quad (16)$$

The corresponding loss profile is

$$\mathcal{L} = (P_0, P_1, \dots, P_D), \quad (17)$$

where

$$P_d = \Pr(\Pi < 0 \mid D = d). \quad (18)$$

This permits direct analysis of whether financial outcomes vary systematically across organizational depth.

7 Recruitment and Downstream Size

Let U_{ij} denote the size of the participant's downstream organization.

A natural empirical model is

$$\Pi_{ij} = \alpha + \beta_1 D_{ij} + \beta_2 \ln(1 + U_{ij}) + \beta_3 R_{ij} + \gamma' X_{ij} + \varepsilon_{ij}. \quad (19)$$

The coefficient on downstream size should not automatically be interpreted causally.

A successful participant may build a larger downline, while a larger downline may increase income.

Thus

$$U_{ij} \rightarrow \Pi_{ij} \quad (20)$$

and

$$\Pi_{ij} \rightarrow U_{ij} \quad (21)$$

may both occur.

8 Rewards and Incentives

Let

$$W_{ij} \quad (22)$$

denote rewards and

$$P_{ij} \quad (23)$$

denote penalties or sanctions.

Possible rewards include commissions, bonuses, rank advancement, recognition, prizes, and access to organizational benefits.

The participant outcome may be represented as

$$\Pi_{ij,t+1} = \alpha_i + \rho \Pi_{ij,t} + \beta W_{ij,t} - \gamma P_{ij,t} + \delta' X_{ij,t} + \varepsilon_{ij,t+1}. \quad (24)$$

This is a dynamic incentive model.

9 Recruitment Incentives

Suppose recruitment produces compensation according to

$$W_{ij} = f(R_{ij}, U_{ij}, S_{ij}), \quad (25)$$

where R is recruitment activity, U downstream size, and S sales activity.

The central empirical question is whether the compensation structure creates incentives that favour recruitment relative to genuine retail activity.

One possible ratio is

$$\mathcal{Q}_{ij} = \frac{\text{Recruitment-Linked Compensation}}{\text{Retail-Sales Compensation}}. \quad (26)$$

The ratio is descriptive and should not itself be interpreted as evidence of improper conduct.

10 Organizational Psychology

MLM organizations provide an unusually rich setting for studying motivation, social identity, leadership, peer effects, commitment, expectations, and organizational climate.

A participant's decision to remain in an organization need not depend exclusively on financial returns.

Let utility be

$$U_{ij} = \Pi_{ij} + N_{ij} + S_{ij} + E_{ij} - C_{ij}, \quad (27)$$

where N represents social benefits, S status or recognition, E expectations of future returns, and C costs.

Thus

$$\Pi_{ij} < 0 \quad (28)$$

does not logically imply

$$U_{ij} < 0. \quad (29)$$

This distinction is important when interpreting persistence.

11 Persistence and Attrition

Let

$$A_{ij,t} = \begin{cases} 1, & \text{if participant } ij \text{ exits at } t, \\ 0, & \text{otherwise.} \end{cases} \quad (30)$$

A discrete-time hazard model can be written as

$$\Pr(A_{ij,t} = 1) = F(\alpha + \beta\Pi_{ij,t} + \gamma D_{ij,t} + \delta W_{ij,t} + \theta' X_{ij,t}). \quad (31)$$

This allows us to ask whether financial losses predict subsequent exit.

However, attrition can also generate survivorship bias.

If low-performing participants exit rapidly, a dataset containing only active participants may substantially overstate the apparent performance of the organization.

12 Survivorship Bias

Let $S = 1$ indicate that a participant remains in the organization.

Then the observed distribution is

$$F_{\text{observed}}(\pi) = F(\pi \mid S = 1). \quad (32)$$

But the population distribution is

$$F_{\text{population}}(\pi) = F(\pi). \quad (33)$$

In general,

$$F(\pi \mid S = 1) \neq F(\pi). \quad (34)$$

Therefore a valid empirical study must attempt to observe both active and former participants.

13 Selection into the Organization

Participation itself is endogenous.

Let

$$Z_{ij} = 1 \quad (35)$$

indicate joining the organization.

Then

$$\Pr(Z_{ij} = 1) = F(X'_{ij}\gamma). \quad (36)$$

Individuals may differ systematically in entrepreneurial ability, wealth, risk tolerance, social networks, expectations, education, and prior experience.

Consequently, the observed participants are not necessarily representative of the general population.

14 A Causal Graph

The principal causal structure can be represented schematically.

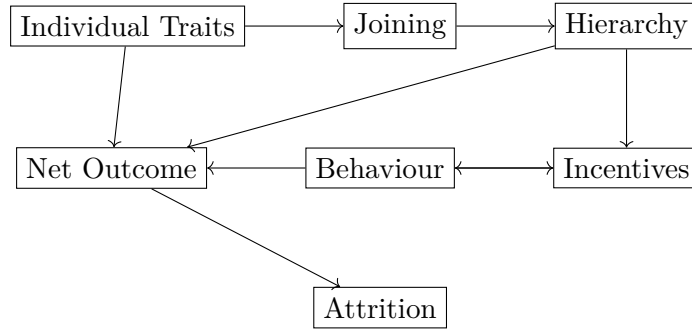


Figure 1: A stylized causal structure for multi-level organizational outcomes.

The diagram emphasizes that incentives and outcomes can influence one another.

15 Econometric Identification

A simple fixed-effects model is

$$\Pi_{it} = \alpha_i + \lambda_t + \beta W_{it} + \gamma P_{it} + \delta' X_{it} + \varepsilon_{it}. \quad (37)$$

Here α_i controls for time-invariant participant or regiment characteristics and λ_t captures common time effects.

However, fixed effects do not automatically eliminate time-varying confounding.

Potential strategies include:

- randomized incentive experiments;
- natural experiments;
- difference-in-differences;
- instrumental variables;

- regression discontinuity designs;
- event studies;
- longitudinal fixed effects;
- matching and weighting;
- sensitivity analysis.

The choice should follow the institutional setting and identifying assumptions.

16 Difference-in-Differences

Suppose an incentive policy affects one group but not another.

A standard specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta(Treat_i \times Post_t) + \varepsilon_{it}. \quad (38)$$

The parameter β measures the treatment effect under the parallel-trends assumption.

The assumption should be examined empirically rather than treated as automatic.

17 Distributional Analysis

Mean outcomes are insufficient when returns are highly skewed.

We therefore study

$$F_{\Pi}(\pi). \quad (39)$$

Useful statistics include:

$$G_{\Pi} = \text{Gini}(\Pi), \quad (40)$$

and

$$Q_{\alpha} = F_{\Pi}^{-1}(\alpha). \quad (41)$$

Quantile regression can estimate

$$Q_{\Pi}(\tau | X) = X' \beta_{\tau}. \quad (42)$$

This allows the effect of organizational position or incentives to differ between low-, median-, and high-performing participants.

18 Expected Value and the 99 Percent Problem

Consider the simple distribution

$$\Pi = \begin{cases} -L, & \text{with probability } p, \\ G, & \text{with probability } 1 - p. \end{cases} \quad (43)$$

Then

$$E[\Pi] = -pL + (1 - p)G. \quad (44)$$

With

$$p = 0.99, \quad (45)$$

we obtain

$$E[\Pi] = -0.99L + 0.01G. \quad (46)$$

Therefore

$$E[\Pi] > 0 \quad (47)$$

if and only if

$$G > 99L. \quad (48)$$

The example demonstrates why a loss percentage cannot substitute for a complete distributional analysis.

19 Financial Economics

The relevant quantity for a participant is net return rather than gross compensation.

Define

$$R_{ij} = \frac{\Pi_{ij}}{K_{ij}}, \quad (49)$$

where K_{ij} represents capital or total committed expenditure.

Time-adjusted performance may be represented by

$$R_{ij}^{(a)} = (1 + R_{ij})^{1/T_{ij}} - 1. \quad (50)$$

Where the data permit, the analysis should distinguish monetary return from compensation, and compensation from economic profit.

20 Opportunity Cost

A participant's true economic cost may include the value of time.

Let H_{ij} denote hours worked and w_{ij}^* the relevant opportunity wage.

Then

$$OC_{ij} = H_{ij}w_{ij}^*. \quad (51)$$

Economic profit becomes

$$\Pi_{ij}^* = \Pi_{ij} - OC_{ij}. \quad (52)$$

This distinction may materially change the measured profitability of participation.

21 Welfare Economics

Private profitability and social welfare are distinct.

Let

$$\mathcal{W} = \sum_i \sum_j [\Pi_{ij} + B_{ij} - C_{ij}], \quad (53)$$

where B includes non-monetary benefits and C includes social costs.

A complete welfare analysis should consider consumer surplus, producer surplus, participant welfare, externalities, information effects, and opportunity costs where measurable.

The objective is not to classify an organizational form as socially good or bad from a single statistic, but to identify measurable welfare consequences.

22 Game Theory

Participants may strategically choose recruitment, sales, effort, and exit.

Let participant j 's action vector be

$$a_j = (e_j, r_j, s_j). \quad (54)$$

The resulting payoff is

$$U_j = U_j(a_j, a_{-j}, H_j). \quad (55)$$

The organizational hierarchy therefore creates a game in which one participant's actions can affect the incentives of others.

Equilibrium behaviour may differ substantially from the behaviour predicted by an isolated-agent model.

23 Network Analysis

The organizational structure can be represented by an adjacency matrix A :

$$A_{ij} = \begin{cases} 1, & \text{if participant } j \text{ is directly connected to } i, \\ 0, & \text{otherwise.} \end{cases} \quad (56)$$

Network statistics may include degree, centrality, clustering, depth, branching factor, and community structure.

The financial outcome can then be modelled as

$$\Pi_t = \rho A \Pi_t + X_t \beta + \varepsilon_t. \quad (57)$$

This allows peer effects and network spillovers to be studied directly.

24 Physics, Chemistry, and Biology

The framework has broader mathematical analogues.

In statistical physics, a large microscopic system can be represented through mesoscopic clusters:

$$\text{microstate} \rightarrow \text{mesoscopic state} \rightarrow \text{macroscopic state}. \quad (58)$$

In chemistry, molecular populations may be grouped according to structural or energetic characteristics.

In biology, individuals may be nested within colonies, populations, ecosystems, or other collective structures.

The analogy is methodological: Regiment Analysis studies how properties of an intermediate grouping relate to individual and aggregate outcomes.

25 Psychology and Psychiatry

Psychological persistence can be represented as a latent utility problem.

Let

$$U_{ijt} = \Pi_{ijt} + S_{ijt} + N_{ijt} + E_{ijt} - C_{ijt}. \quad (59)$$

A participant may remain despite financial losses when non-financial utility or expected future returns compensate for current losses.

In longitudinal psychological data, one may therefore model

$$\Pr(A_{ijt} = 1) = F(-U_{ijt}, X_{ijt}). \quad (60)$$

Clinical applications require substantially stronger ethical and methodological safeguards and should not infer mental-health conditions from organizational participation.

26 Political Science and International Relations

The same mathematical architecture applies to political organizations, parties, lobbying networks, international institutions, alliances, and diplomatic coalitions.

A coalition can be represented as

$$C_i = \{s_1, s_2, \dots, s_q\}, \quad (61)$$

where s_j are states or political actors.

Rewards may include economic assistance, institutional access, diplomatic recognition, or cooperation; sanctions may include exclusion or economic penalties.

The central statistical problem remains selection and strategic interaction.

27 Geopolitics and Diplomacy

International organizations can exhibit hierarchical and networked structures without being MLM organizations.

The framework therefore transfers only at the mathematical level.

For example,

$$C_{i,t+1} = F(C_{it}, W_{it}, P_{it}, X_{it}) \quad (62)$$

can describe the evolution of cooperation within a diplomatic grouping.

The model does not imply that states behave like commercial participants.

28 Warfare Economics

Historical and abstract military organizations also contain nested units, incentives, rewards, sanctions, promotion systems, and attrition.

The empirical variables might include organizational performance, personnel retention, resource allocation, and administrative outcomes.

No operational battlefield prescription follows from the model.

The mathematical object remains

$$Y_{i,t+1} = F(Y_{it}, W_{it}, P_{it}, X_{it}). \quad (63)$$

29 Computer Science and Data Architecture

A raw organizational dataset can be transformed into a regiment-level dataset:

$$D_{\text{raw}} \rightarrow D_{\text{participant}} \rightarrow D_{\text{regiment}} \rightarrow D_{\text{network}} \rightarrow D_{\text{model}}. \quad (64)$$

This architecture supports scalable processing.

Within-regiment statistics can be computed independently:

$$\Phi_i = f(R_i), \quad (65)$$

after which the resulting profiles can be processed centrally.

The procedure is therefore naturally parallelizable.

30 Algorithm

A practical empirical pipeline is:

MULTILEVEL_REGIMENT_ANALYSIS(D) :

```
CLEAN(D)
DEFINE_NET_OUTCOME(D)
RECONSTRUCT_HIERARCHY(D)
IDENTIFY_REGIMENTS(D)

for each regiment i:
    COMPUTE_PROFILE(i)
    COMPUTE_DEPTH(i)
    COMPUTE_RECRUITMENT(i)
    COMPUTE_REWARDS(i)
    COMPUTE_PUNISHMENTS(i)
    COMPUTE_ATTRITION(i)

ESTIMATE_OUTCOME_DISTRIBUTION()
ESTIMATE_MULTILEVEL_MODELS()
TEST_SURVIVORSHIP_BIAS()
TEST_REVERSE_CAUSALITY()
ESTIMATE_CAUSAL_EFFECTS_WHERE_IDENTIFIED()
ESTIMATE_NETWORK_EFFECTS()
RUN_ROBUSTNESS_TESTS()
RETURN_RESULTS()
```

31 Machine Learning

The regiment profile can be represented as

$$\Phi_i = (D_i, U_i, R_i, W_i, P_i, \bar{\Pi}_i, \sigma_{\Pi,i}, A_i, \dots). \quad (66)$$

Machine-learning algorithms can then estimate

$$\hat{A}_i = f_\theta(\Phi_i), \quad (67)$$

where $\hat{A}_i$ is predicted attrition.

Possible algorithms include random forests, gradient boosting, support-vector machines, neural networks, clustering, survival models, and graph neural networks.

The predictive model should not be confused with a causal model.

32 Anomaly Detection

A regiment can be identified as anomalous when its profile deviates substantially from expected profiles.

Define

$$Z_i = (\Phi_i - \mu_\Phi)' \Sigma_\Phi^{-1} (\Phi_i - \mu_\Phi). \quad (68)$$

Large Z_i values identify multivariate outliers under the assumptions of the chosen model.

This can identify unusual recruitment structures, unusually concentrated rewards, exceptional attrition, or anomalous financial outcomes.

33 Robustness

A credible empirical analysis should vary:

- the definition of profit;
- the treatment of expenses;
- the time horizon;
- the participant denominator;
- treatment of inactive participants;
- treatment of missing observations;
- hierarchy reconstruction;
- outlier rules;
- inflation adjustment;
- opportunity-cost assumptions.

The principal result should survive reasonable alternative definitions.

34 Data Requirements

An ideal dataset would contain:

Table 1: Minimum empirical data architecture

Variable	Required information
Participant ID	Unique longitudinal identifier
Parent ID	Immediate organizational superior
Join date	Entry timing
Exit date	Attrition timing
Hierarchy depth	Organizational position
Recruitment	Number and timing of recruits
Downline size	Downstream network
Sales	Gross and net sales
Commissions	Compensation received
Bonuses	Additional rewards
Fees	Organizational charges
Purchases	Participant expenditure
Hours	Time commitment
Expenses	Other operating costs
Training	Training-related costs
Performance	Relevant output measure

The strongest design is longitudinal because it permits temporal ordering.

35 Empirical Tests

The principal hypotheses can be summarized as follows.

Table 2: Illustrative empirical hypotheses

Hypothesis	Test	Main quantity
H1	Depth-performance relation	$E[\Pi \mid D]$
H2	Recruitment-performance relation	β_R
H3	Reward response	β_W
H4	Punishment response	β_P
H5	Attrition after losses	Hazard ratio
H6	Survivorship bias	$F(\Pi \mid S = 1)$
H7	Network effects	ρ
H8	Outcome concentration	Gini/quantiles
H9	Heterogeneous responses	$\text{Var}(\beta_i)$
H10	Threshold effects	Discontinuity/bunching

36 Propositions

36.1 Proposition 1: Loss Probability Does Not Determine Expected Return

Let

$$\Pr(\Pi = -L) = p \tag{69}$$

and

$$\Pr(\Pi = G) = 1 - p. \quad (70)$$

Then

$$E[\Pi] = -pL + (1 - p)G. \quad (71)$$

Therefore the probability of loss alone does not determine expected return.

Proof. The expectation depends on both the probability and magnitude of each outcome. Hence two distributions can have identical loss probabilities and different expected returns. $\square$

36.2 Proposition 2: Survivorship Can Bias Observed Profitability

Suppose exit probability depends on financial performance:

$$\Pr(S = 1 \mid \Pi) = F(\Pi). \quad (72)$$

Then generally

$$F(\Pi \mid S = 1) \neq F(\Pi). \quad (73)$$

Proof. Conditioning on a variable that depends on the outcome changes the distribution unless selection is independent of the outcome. $\square$

36.3 Proposition 3: Hierarchical Position Does Not Establish Causality

Suppose

$$D_i = f(\theta_i, \nu_i) \quad (74)$$

and

$$\Pi_i = g(D_i, \theta_i, \varepsilon_i). \quad (75)$$

If latent characteristics θ_i affect both hierarchy and profitability, then the observed relationship between D_i and Π_i does not identify the causal effect of hierarchy.

Proof. The common dependence on θ_i generates omitted-variable bias unless θ_i is controlled or otherwise addressed by the identification strategy. $\square$

37 Ethics and Privacy

Organizational datasets can contain sensitive financial, employment, social-network, and behavioural information.

Participant identifiers should therefore be pseudonymized or anonymized where appropriate. The empirical framework should observe:

$$\text{scientific validity} + \text{privacy} + \text{informed consent} + \text{responsible reporting}. \quad (76)$$

Individual-level predictions should not be used to stigmatize participants or infer sensitive personal characteristics without appropriate justification and safeguards.

38 A General Theory of Multi-Level Organizational Outcomes

The framework can be summarized by the following dynamic system:

$$\text{individual characteristics} \rightarrow \text{organizational entry}, \quad (77)$$

$$\text{entry} \rightarrow \text{hierarchical position}, \quad (78)$$

$$\text{hierarchical position} \rightarrow \text{incentives}, \quad (79)$$

$$\text{incentives} \rightarrow \text{behaviour}, \quad (80)$$

$$\text{behaviour} \rightarrow \text{financial outcome}, \quad (81)$$

$$\text{financial outcome} \rightarrow \text{attrition or persistence}, \quad (82)$$

$$\text{attrition} \rightarrow \text{future organizational structure}. \quad (83)$$

Thus,

Entry $\rightarrow$ Hierarchy $\rightarrow$ Incentives $\rightarrow$ Behaviour $\rightarrow$ Outcome $\rightarrow$ Attrition $\rightarrow$ New Hierarchy
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(84)

The organization is therefore a dynamic system rather than a static tree.

39 Conclusion

Multi-level organizations provide an unusually rich environment for empirical Regiment Analysis.

The MLM setting is particularly useful because hierarchy, recruitment, rewards, downstream networks, financial outcomes, and attrition are often directly connected. This makes it possible to study the entire sequence from organizational entry through hierarchy and incentives to performance and exit.

The frequently cited claim that at least 99 percent of recruits lose money is therefore best treated as a motivating empirical proposition rather than as a universal constant. A rigorous study must establish the population, denominator, accounting definition of loss, observation period, treatment of expenses, treatment of inactive participants, and selection mechanism.

The central empirical object is the distribution

$$F_{\Pi}(\pi \mid D, U, R, W, P, X). \quad (85)$$

The central causal question is

$$\frac{\partial \Pi_{t+1}}{\partial (W_t, P_t, D_t, R_t)}. \quad (86)$$

The central organizational question is

<i>How does hierarchical position interact with incentives to produce heterogeneous outcomes and attrition?</i>

(87)

Regiment Analysis therefore provides a framework in which organizational hierarchy is not merely a descriptive tree. It becomes an empirically testable system of selection, incentives, behaviour, performance, network dependence, and survival.

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Glossary

Multi-Level Organization

An organization in which participants occupy multiple nested hierarchical or network positions.

Multi-Level Marketing

A business structure in which participants may receive compensation associated with sales, recruitment, or downstream organizational activity.

Regiment

A mesoscopic organizational grouping used as the principal unit of analysis.

Regiment Analysis

Statistical and econometric analysis of the characteristics, outcomes, incentives, and dynamics of identified regiments.

Hierarchy Depth

The number of organizational levels separating a participant from a designated organizational origin.

Downline

Participants or organizational units located downstream from a participant in a hierarchical network.

Net Outcome

Gross monetary receipts minus relevant economic expenses.

Reward

A positive monetary, organizational, or status incentive.

Punishment

A negative monetary, organizational, or status consequence.

Attrition

Exit of a participant from an organization.

Survivorship Bias

Bias arising when analysis conditions on continued participation or observation.

Selection

The non-random process through which individuals enter, remain in, or occupy positions within an organization.

Incentive Response

The behavioural change associated with a change in rewards or sanctions.

Distributional Outcome

The complete distribution of outcomes rather than only an average or aggregate statistic.

Opportunity Cost

The value of the best alternative use of resources, including time.

Mesoscopic Unit

An analytical unit between an individual observation and the complete system.

Network Effect

An effect transmitted through relationships among organizational participants.

Regiment Profile

A vector summarizing the structural, financial, behavioural, and incentive characteristics of a regiment.

The End